

ERYTHROMYCIN SOLUBLE POWDER

300mg/g

Ingredient(s):

Each g contains :
Erythromycin Thiocyanate 300mg (potency)

Preservative(s):

Each g contains:
Sodium Benzoate.....1 mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):

Erythromycin is usually a bacteriostatic agent, but in high concentrations or against highly susceptible organisms it may be bactericidal. Erythromycin appears to inhibit protein synthesis in susceptible organisms by reversibly binding to 50S ribosomal subunits, thereby inhibiting translocation of aminoacyl transfer-RNA and inhibiting polypeptide synthesis.

Following oral administration, Erythromycin is absorbed rapidly from the upper part of the small intestine and diffuses well throughout most tissues. Serum concentrations decline sharply within 4~6 hours after the drug is administered.

Erythromycin given by IM injection has a plasma half-life of 8~9 hours. 2 hours after administration, highest concentrations are detected in liver, submaxillary glands, lungs, and kidneys. Only trace amounts are present in brain or cerebrospinal fluid.

Diffusion into pleural and peritoneal fluid as well as across the placenta into fetal circulation occurs readily. Erythromycin is present in prostatic fluid and semen in about one-third the concentration of that found in blood. It is also present in nasal mucus. Erythromycin may be 73~81% bound to serum proteins and the estolate salt, 96% bound.

Pharmacokinetic parameters of Erythromycin in calves may change in the presence of *Pasteurella hemolytica* infection. Erythromycin half-life changes from 133 minutes in normal calves to 100~111 minutes in calves with minor and major pneumonic lesions.

Erythromycin is primarily excreted unchanged in the bile, and is also partly metabolized by the liver via N-demethylation to inactive metabolites. Some of the drug is reabsorbed after biliary excretion. Only about 2~5% of a dose is excreted unchanged in the urine. The reported elimination half-life of Erythromycin in various species are: 60~90 minutes in dogs and cats, 60~70 minutes in foals and mares, and 190 minutes in cattle.

Indication(s):

As an aid in the prevention and reduction of lesions and in lowering the severity of poultry chronic respiratory disease during period of stress in chickens and turkeys, Bluecom, Infectious coryza, Sinovitis, Sinusitis and Staphylococcosis.

Dosage and Administration:

Respiratory diseases and complex disease syndromes: Dissolve 38.5g into 100L of drinking water.

Stresses: Dissolve 19.25g into 100L of drinking water.

Use as the only source of water for 5 days. Continue medication for 2 more days if the birds have not fully responded.

Contraindications:

- 1) Contraindicated in patients hypersensitive to erythromycin.
- 2) The estolate form should probably be avoided in patients with preexisting liver dysfunction.
- 3) As it may induce a toxic enterocolitis, erythromycin is contraindicated in rabbits, gerbils, guinea pigs and hamsters.
- 4) It is believed that erythromycin is contraindicated in adult horses, and oral erythromycin should not be used in ruminants as severe diarrhea may result.

Precaution(s) / Warning(s):

- 1) Oral erythromycin should not be used in ruminants as severe diarrheas may result.
- 2) In foals treated with erythromycin, a mild, self-limiting diarrhea may occasionally occur. Adult horses may develop severe, sometimes fatal diarrheas from erythromycin and the use of the drug in adult horses is very controversial.
- 3) Erythromycin may alter temperature homeostasis in foals. Foals between the ages of 2 and 4 months old have been reported to develop hyperthermia with associated respiratory distress and tachypnea.

Interaction with Other Medicaments:

- 1) Because erythromycin, lincosamides (clindamycin, lincomycin), and chloramphenicol all bind to the 50S ribosomal subunit, competition for binding can occur and these drugs should not be used concurrently.
- 2) *In vitro* synergy with other antimicrobials (e.g., sulfonamides, rifampin) has been reported with erythromycin.
- 3) The concomitant use of erythromycin with bactericidal antibiotics (e.g., penicillin) is controversial, but documentation of either clinical synergy, additive activity, or antagonism is apparently lacking.
- 4) Decreased clearance of theophylline may occur with resultant toxicity in patients receiving erythromycin (particularly high dosages).
- 5) Patients stabilized on warfarin anticoagulant therapy may develop prolonged prothrombin times and bleeding when erythromycin is added.
- 6) The metabolism of methylprednisolone may be inhibited by concurrent administration of erythromycin.
- 7) Erythromycin may increase the bioavailability of digoxin in a small percentage of human patients and can lead to digoxin toxicity.
- 8) Interactions have also been reported with erythromycin and the following human drugs: carbamazepine, cyclosporine, and triazolam.

Pregnancy and Lactation:

While erythromycin has not demonstrated teratogenic effects in rats and the drug is not thought to possess serious teratogenic potential, it should only be used during pregnancy when the benefits outweigh the risks.

Side Effect(s) / Adverse Reaction(s):

- 1) Adverse effects are relatively infrequent with erythromycin when used in small animals, swine, sheep, or cattle.
- 2) Oral erythromycin may cause GI disturbances with diarrhea, anorexia, and vomiting occasionally seen.
- 3) Rectal edema and partial anal prolapse have been associated with erythromycin in swine.
- 4) When used as prokinetic agent, erythromycin may actually increase symptoms of intestinal distress as it can stimulate emptying of larger food particles into the intestine than is normal.

Environmental property:

None has been mentioned in animal.

Symptoms and Treatment for Overdosage, and Antidote(s):

In foals develop hyperthermia, physically cooling off these animals is reported to be successful in controlling this effect.

Shelf-Life:

4 years.

Reconstituted preparation should be used up within 24 hours

Storage Condition(s):

Store at temperature below 30°C.

Product Description and Packing(s):

A white color powder and without other foreign matter.

Al. Pouch of 75g, 150g & 500g

For food producing animals product:**Withdrawal Periods**

Chicken & turkey, meat: 2 days before slaughter.

Egg: Do not use in birds producing eggs for human consumption.

Maximum Residual Limit (MRL):

Chicken & turkey meat, fat, liver & kidney 200µg/kg

Egg 15µg/kg

**Y. S. P.**

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